

STELLA TECHNOLOGY — INTERNSHIP FINAL PROJECT

API Reference

OpenAPI 3.0 specification summary — all endpoints, schemas, and response codes

Base URL: `http://localhost:8080` **Auth:** JWT Bearer Token

Spec file: `openapi.yaml` (importable into Swagger Editor & Postman)

The full machine-readable spec is in `openapi.yaml`. Import it into [Swagger Editor](#) ([editor.swagger.io](#) → File → Import File) for interactive documentation, or into [Postman](#) (Import → select file) to generate a ready-to-use request collection.

Authentication

All endpoints except `POST /api/auth/login` require a **JWT Bearer token** in the Authorization header:

```
Authorization: Bearer <token>
```

Tokens are issued on login and expire after **24 hours**.

Role Legend

Badge	Meaning
ADMIN	Only admin accounts can call this endpoint
HR	HR accounts — data is auto-scoped to their department
EMPLOYEE	Any authenticated employee
ALL	Any authenticated role

Auth Endpoints

Method	Path	Role	Description
POST	/api/auth/login	Public	Authenticate and receive a JWT token
GET	/api/auth/me	ALL	Get current user's email and role

POST /api/auth/login — Request

Field	Type	Required	Notes
email	string	Yes	Case-insensitive
password	string	Yes	

POST /api/auth/login — Response 200

Field	Type	Description
token	string	Signed JWT to use as Bearer token
role	string	ADMIN HR EMPLOYEE
email	string	The authenticated email

Employee Endpoints

Method	Path	Role	Description
GET	/api/employees	ADMIN HR	Paginated list — supports name, status, userRole, departmentId filters
POST	/api/employees	ADMIN	Register a new employee (creates user account + profile)
GET	/api/employees/{employeeId}	ADMIN HR	Get single employee profile
PUT	/api/employees/{employeeId}	ADMIN	Update employee details

Method	Path	Role	Description
DELETE	/api/employees/{employeeId}	ADMIN	Delete employee and all associated data
GET	/api/employees/me	ALL	Current user's own profile (with departmentId)
GET	/api/employees/{employeeId}/history	ADMIN EMPLOYEE HR	Promotion, designation change, and note history
POST	/api/employees/{employeeId}/notes	ADMIN	Add a performance note

POST /api/employees — Request Body

Field	Type	Required	Notes
name	string	Yes	
email	string	Yes	Must be unique
password	string	Yes	Min 6 characters
phone	string	Yes	
gender	enum	Yes	MALE FEMALE
dateOfBirth	date	Yes	Format: YYYY-MM-DD
address	string	Yes	
departmentId	string	No (HR)	Required for EMPLOYEE/MANAGER. HR defaults to DEPT-0000.
designation	string	Yes	Job title
dateOfJoining	date	Yes	Format: YYYY-MM-DD
salary	number	Yes	Base monthly salary
allowances	number	No	Default: 0
deductions	number	No	Default: 0
employeeType	enum	Yes	EMPLOYEE MANAGER HR
userRole	enum	Yes	EMPLOYEE HR

Department Endpoints

Method	Path	Role	Description
GET	/api/departments	ADMIN HR	List all departments
POST	/api/departments	ADMIN	Create a department (hrId required)
PUT	/api/departments/{departmentId}	ADMIN	Update department details
DELETE	/api/departments/{departmentId}	ADMIN	Delete department; use ?targetDepartmentId= to reassign employees

POST /api/departments — Request Body

Field	Type	Required	Notes
name	string	Yes	Must be unique
description	string	No	
hrId	string	Yes	Must reference an employee with userRole HR. One HR per department.
managerId	string	No	Must reference an employee with employeeType MANAGER

Leave Endpoints

Method	Path	Role	Description
POST	/api/leave/apply	HR EMPLOYEE	Submit a leave request
GET	/api/leave/my	HR EMPLOYEE	Own leave request history
GET	/api/leave/my-balance	ALL	Own remaining leave quotas
GET	/api/leave/all	ADMIN	All leave requests org-wide
GET	/api/leave/department	HR	Leave requests for HR's department (auto-scoped)

Method	Path	Role	Description
POST	/api/leave/{leaveld}/approve	HR	Approve a PENDING request; decrements balance
POST	/api/leave/{leaveld}/reject	HR	Reject a PENDING request
GET	/api/leave/balances	ADMIN HR	Balance lookup — optional ?name= filter

POST /api/leave/apply — Request Body

Field	Type	Required	Notes
leaveType	enum	Yes	ANNUAL CASUAL SICK UNPAID
startDate	date	Yes	YYYY-MM-DD. Must not be after endDate.
endDate	date	Yes	YYYY-MM-DD
reason	string	Yes	

Leave Balance Object

Field	Type	Default
annualLeaves	number	18
casualLeaves	number	10
sickLeaves	number	8
unpaidLeaves	number	5

Payroll Endpoints

Method	Path	Role	Description
POST	/api/payroll/generate	HR	Generate monthly payroll (dept employees only)
PUT	/api/payroll/{payrollId}/status	HR	Update payment status (PAID / UNPAID)
GET	/api/payroll/all	ADMIN	All payroll org-wide; filter by month, year, dept, status

Method	Path	Role	Description
GET	/api/payroll/department	HR	Payroll for HR's department (auto-scoped)
GET	/api/payroll/my	ALL	Own payroll history

POST /api/payroll/generate — Request Body

Field	Type	Required	Notes
employeeId	string	Yes	Must be in HR's department
month	number	Yes	1–12
year	number	Yes	e.g. 2026

Payroll Response Object

Field	Type	Notes
payrollId	string	Sequential: PAY-0001, PAY-0002, ...
employeeId	string	
month	number	1–12
year	number	
basicSalary	number	Copied from employee at generation time
allowances	number	
deductions	number	
netSalary	number	basicSalary + allowances – deductions
paymentStatus	enum	UNPAID PAID
generatedAt	datetime	ISO 8601

Error Response Format

All error responses follow the same structure:

```
{
  "status": 400,
```

```
"error": "Bad Request",
"message": "Payroll already exists for this employee in September 2026",
"path": "/api/payroll/generate"
}
```

HTTP Status Codes

Code	Description	Common Causes
200 OK	Success	GET, PUT, DELETE succeeded normally
201 Created	New resource created	POST to register/apply/generate
400 Bad Request	Validation failed	Missing field, duplicate, insufficient leave balance, cross-dept payroll
401 Unauthorized	Authentication missing or invalid	No token, expired token, wrong token
403 Forbidden	Authenticated but not allowed	EMPLOYEE calling ADMIN-only endpoint
404 Not Found	Resource does not exist	Bad employeeId, leaveId, payrollId, departmentId

Viewing the Full Spec

The `openapi.yaml` file in the `submission/` folder is a complete OpenAPI 3.0.3 specification. To view it interactively:

Option A — Swagger Editor (no install)

1. Open a browser and go to editor.swagger.io
2. Click **File** → **Import File**
3. Select `submission/openapi.yaml`
4. The right panel shows the full interactive documentation
5. You can try requests directly from the browser (requires CORS-enabled backend)

Option B — Postman

1. Open Postman → click **Import** (top-left)
2. Select `submission/openapi.yaml`

3. Postman generates a full collection from the spec
4. Set the base URL and JWT token in the collection variables

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